

Hyperparameter tuning and model validation – Lesson overview

Hyperparameters set the stage for **how machine learning models learn**, significantly affecting their performance and accuracy. Model validation then **measures this performance**, ensuring models generalise well to new data. Together, these steps are crucial for creating robust machine learning applications.

In this lesson, we'll explore hyperparameter tuning, emphasising **grid search**, a methodical approach to **finding the best model settings**. We'll also delve into model validation techniques, learning how to assess model performance accurately.



Video



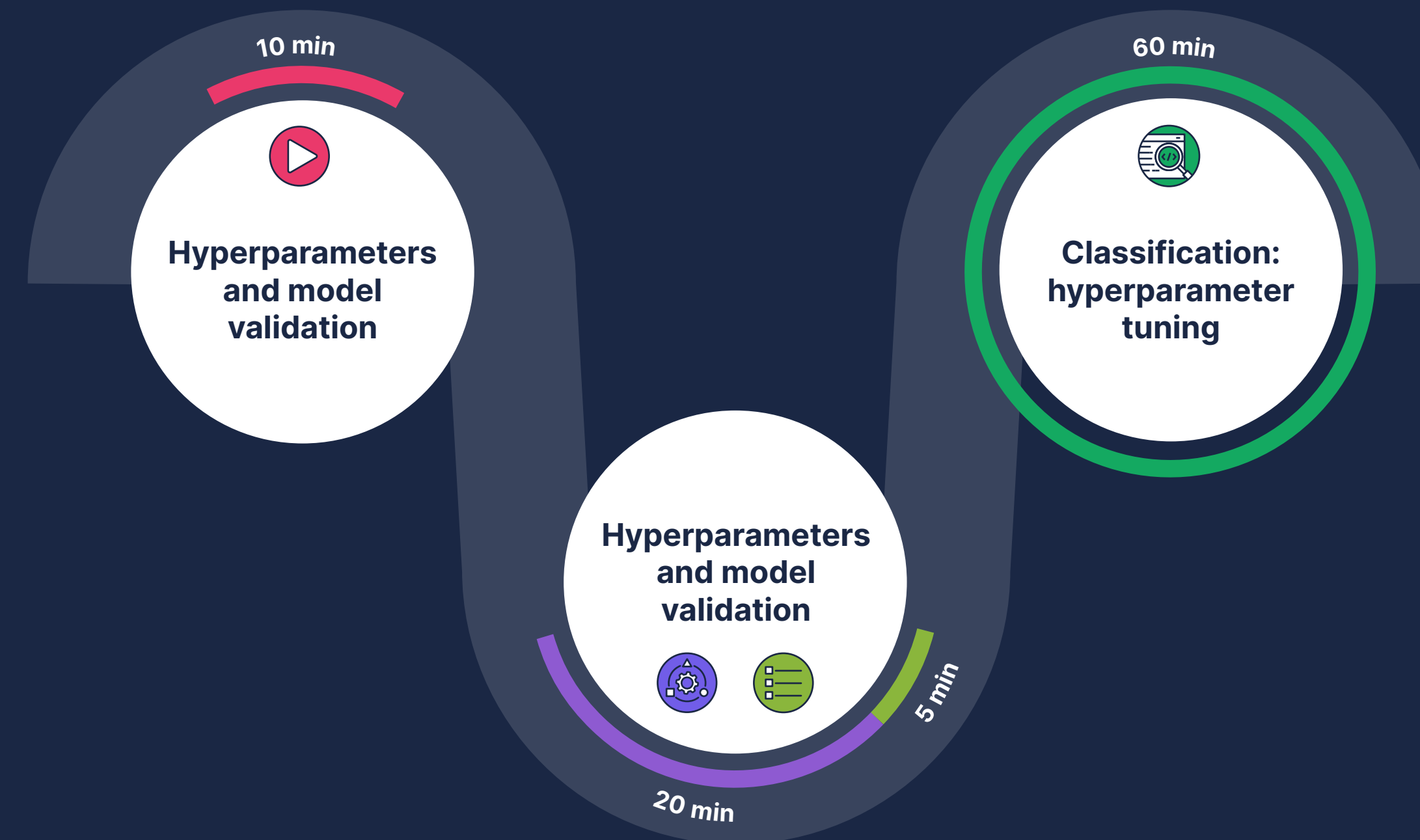
Knowledge questions



Examples



Code challenge



Learning objectives

- Define and differentiate between model parameters and hyperparameters.
- Implement grid search for hyperparameter tuning.
- Understand and apply basic model validation techniques to assess model performance.
- Apply hyperparameter tuning strategies to improve classification models.

